INTRODUCTION

DECIMAL → HEXADECIMAL

Hexadecimal is base 16.

Convert by repeatedly dividing by 16.

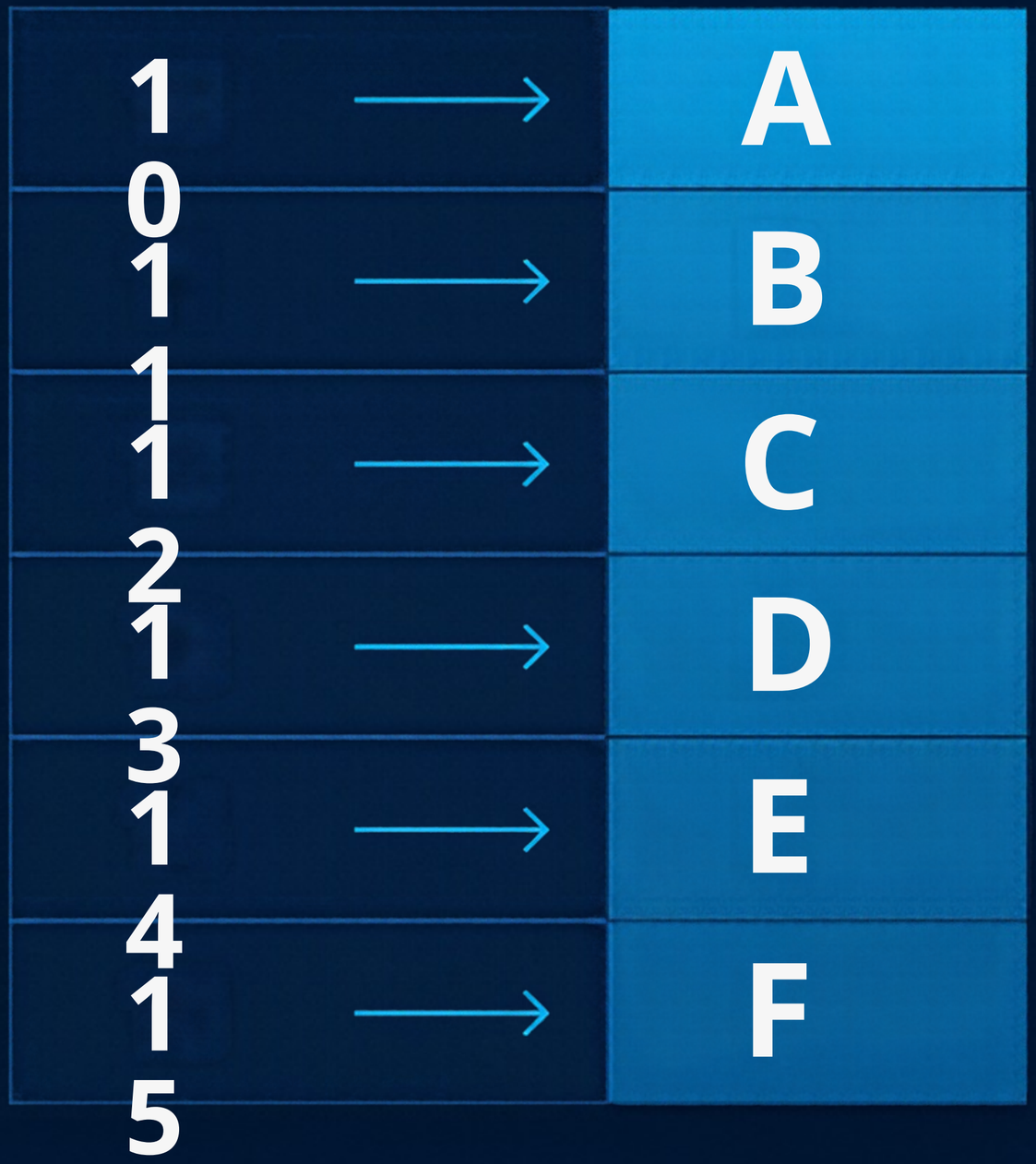
Remainders can be 0-9 or letters A-F.

Letters A-F represent remainders 10-15.



HEXADECIMAL VALUE FOR 10 TO 15

These are the hexadecimal digit equivalents for decimal 10 to 15.



WORKED EXAMPLE

Convert 254_{10} to Hexadecimal

254 ÷ 16 = 15	remainder 14	→ 14 → E
15 ÷ 16 = 0	remainder 15	→ 15 → F

Read remainders bottom to top

Therefore,

$$254_{10} = FE_{16}$$

PRACTICE EXAMPLE

YOUR TURN!

Convert 26_{10}---> ?16

2 SECONDS

SOLUTION

26 ÷ 16 = 1 remainder **10**

(10 → **A**)

↓

1 ÷ **16** = 0 remainder **1**

↓

Read bottom to top: **1A**

Therefore, **2610 = 1A16**

RECAP AND NEXT LESSON

01 Divide decimal number by 16 repeatedly.

02 Record the remainders.

03 Convert remainders 10–15 into letters A–F.

04 Read the remainders from bottom to top.

NEXT LESSON:

HEXADECIMAL TO DECIMAL

We’ll convert hexadecimal back to decimal.